Pharma Cold Chain Data Warehouse & Analytics Platform

Phase 1, Milestone 1: Progress Report (Updated)

Course: Intelligent Data Architectures | Prepared by: Narin Nadeem | Date: 11 November 2026

1. Executive Summary

Phase 1 builds the Sales and Demand Intelligence (SDI) mart from three source files: master data (63 medicines), pharmacy sales (500,000 rows), and hospital demand (27,216 rows). Milestone 1 establishes the automated data engineering pipeline from raw CSV landing to the analytical query layer, orchestrated by Apache Airflow 3.4.0.

Component	Status & Verification
Environment (Docker, Airflow 3.4.0, PostgreSQL 16, DuckDB)	**Done** — Operational under WSL 2 with 5GB RAM allocation.
Synthetic Data Generator (`oversample.py`)	**Done** — Preserves ID continuity into 2026 batches across landing files.
Ingestion & Validation (PostgreSQL Staging)	**Done** — Loaded 520,000+ sales and 28,728+ hospital records; reject table active.
Star Schema (DuckDB with SCD Type 2 on `Dim_Medicine`)	**Done** — Fact constellation tables populated; SCD Type 2 track rows verified.
Analytical Query Layer	**Done** — 3 business queries return results within 13 ms to 16 ms.
Pipeline Observability	**Done** — Metrics logged to `staging.pipeline_metrics_batch`.
Airflow DAG End-to-End Orchestration	**Done** — 100% green success on both CLI tests and Web UI Celery execution.
File-watcher Trigger	**In Progress** — Scheduled for upcoming milestone.

2. Architecture & Data Pipeline Flow

Stage	Operational Responsibility	Tool / Script
Generate	Oversamples historical patterns, generating timestamped incremental batches.	`oversample.py`
Ingest & Validate	Validates records, flags rejects to dead-letter table, bulk COPY loads to PostgreSQL.	`ingest_staging.py` (PostgreSQL 16)
Model	Transforms staging tables into dimensional star schema with SCD Type 2 tracking.	`build_datamart.py` (DuckDB)
Query	Executes dimensional analytical SQL queries and captures runtime performance metrics.	`run_queries.py`
Orchestrate	Coordinates 4 dependent sequential tasks, managing task state and lineage.	Apache Airflow (CeleryExecutor)

3. Data Generation Logic & Ingestion Metrics

The synthetic data generator samples historical distributions, reading the latest maximum record ID from both `data/raw` and `data/landing` to guarantee unique continuous numbering. Dates advance monotonically into 2026. Staging maintains file load idempotency via `staging.pipeline_metrics_batch` tracking:

Source File	Rows Loaded	Execution Duration
01_master_data.csv	63	0.10 s
05_pharmacy_sales_data.csv	500,000	28.57 s
incremental_sales_batch_01.csv	25,000	1.10 s
06_hospital_demand_data.csv	27,216	0.93 s
incremental_hosp_batch_01.csv	2,721	0.11 s

4. Orchestration Status & UI Execution Verification

The DAG `phase1_sdi_pipeline` coordinates four sequential tasks: `synthesize_incremental_batches` → `stage_and_validate_postgres` → `transform_duckdb_star_schema` → `run_analytical_and_storage_queries`.
Status: Successfully verified end-to-end via Web UI manual trigger (Run ID: `manual__2026-11-11T01:15:34.735318+00:00` by user `admin`). Total pipeline run duration was **3m 22s** with zero task failures. Task durations: synthesis (25.3s), ingestion/validation (4.8s), DuckDB star schema transformation (2m 39s), and analytical query evaluation (4.35s).

5. Infrastructure Issues Resolved

Symptom	Root Cause	Resolution
UI Task Rejection (`Invalid auth token`)	Mismatched dynamic JWT secrets between Airflow worker and API server containers.	**Fixed**: Configured static `AIRFLOW__API_AUTH__JWT_SECRET` in Compose.
Container SIGBUS Crash	Host disk exhaustion (0.2 GB remaining on C: drive).	**Fixed**: Relocated Docker storage image to secondary drive.
DuckDB Transform OOM Kill	Default Docker memory limit (3.65 GB) exceeded.	**Fixed**: Set `memory=5GB` in `.wslconfig`.
ID Collisions on Incremental Loads	Generator queried raw tables without checking landing directory max IDs.	**Fixed**: Scan both raw and landing directories before generating keys.

6. Remaining Phase 1 Deliverables

1. Capture second DAG refresh run to demonstrate dynamic data mart growth. 2. Finalize the 1–2 page Data Generation Logic report. 3. Prepare BI dashboard integration (Milestone 2) for operational ingestion and query metrics.